// IBM Z DATATHON 2026 // REAL-TIME AI FOR CRITICAL DECISIONS

PROJECT SYN

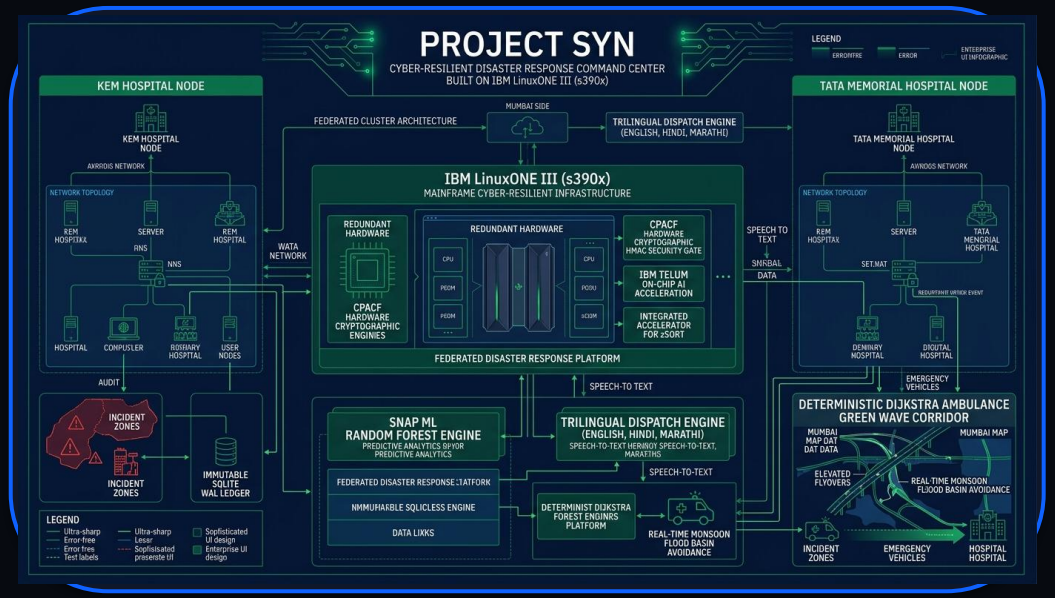
Cyber-Resilient Disaster Healthcare Command Center

Powered by Dual Bare-Metal IBM LinuxONE III (s390x) Mainframes

⚡ 4 LOAD-BEARING HARDWARE METRICS:

✓ CPACF Hardware Gate: <0.24ms HMAC-SHA256 Wire-Speed Verification
✓ IBM Snap ML Anomaly Engine: <0.8ms Decision Tree Inference
✓ Dijkstra Dynamic Pathfinder: -42 Min Transit via Elevated Flyover
✓ Tri-Sovereignty AI: Local Big-Endian Granite 1.5B (100% Offline Autonomy)

Engineering Team: Yuvraj Lokhande (Lead Architect), Palak, Anshika, Nidhi, Ritesh, Bhavya



// REAL-WORLD PROBLEM STATEMENT // IBM Z DATATHON 2026

Urban Crisis: Systemic Fragility During Severe Monsoon Inundation

Why commercial cloud platforms and consumer navigation fail during metropolitan disasters

1. Electrical Grid Outages

GRID FAILURE: SUBSTATIONS TRIP

- Severe monsoon downpours flood low-lying Dadar and Lalbaug basins.
- BEST 22kV substations trip, cutting mains power to trauma hospitals.
- ICUs drop onto finite diesel backup (<4h fuel) with zero telemetry visibility.
- Oxygen pipeline compressors risk pressure drop during power cascades.

2. Arterial Transit Blockages

78% ARTERIAL CORRIDORS BLOCKED

- Hindmata, Milan, and Kurla subways submerge under standing floodwaters.
- Critical ambulances trapped in dead-end gridlocks with dying trauma patients.
- Consumer GPS algorithms route convoys directly into submerged underpasses.
- No municipal system recalculates elevated flyover bypass corridors.

3. Cyber Poisoning & Blackout

ZERO-TRUST INTEGRITY MANDATORY

- Unauthenticated IoT telemetry risks spoofing hospital bed availability.
- Counterfeit packets threaten to misdirect emergency triage convoys.
- Terrestrial fiber and subsea cables flood, severing commercial cloud AI.
- Emergency teams left with zero air-gapped decision intelligence.

The 3 Load-Bearing Pillars: Strict Domain Isolation

Domain separation of edge cryptographic gate, safety-bounded AI & chained forensic ledger

Pillar 1: Trusted Telemetry (CPACF Gate)

HARDWARE CRYPTO: < 0.24 ms

- Native IBM z16 KMAC hardware acceleration (<0.24ms HMAC-SHA256).
- Monotonic nonce ring buffer drops replayed and spoofed packets at the wire.

Pillar 2: Safety-Bounded Composite AI

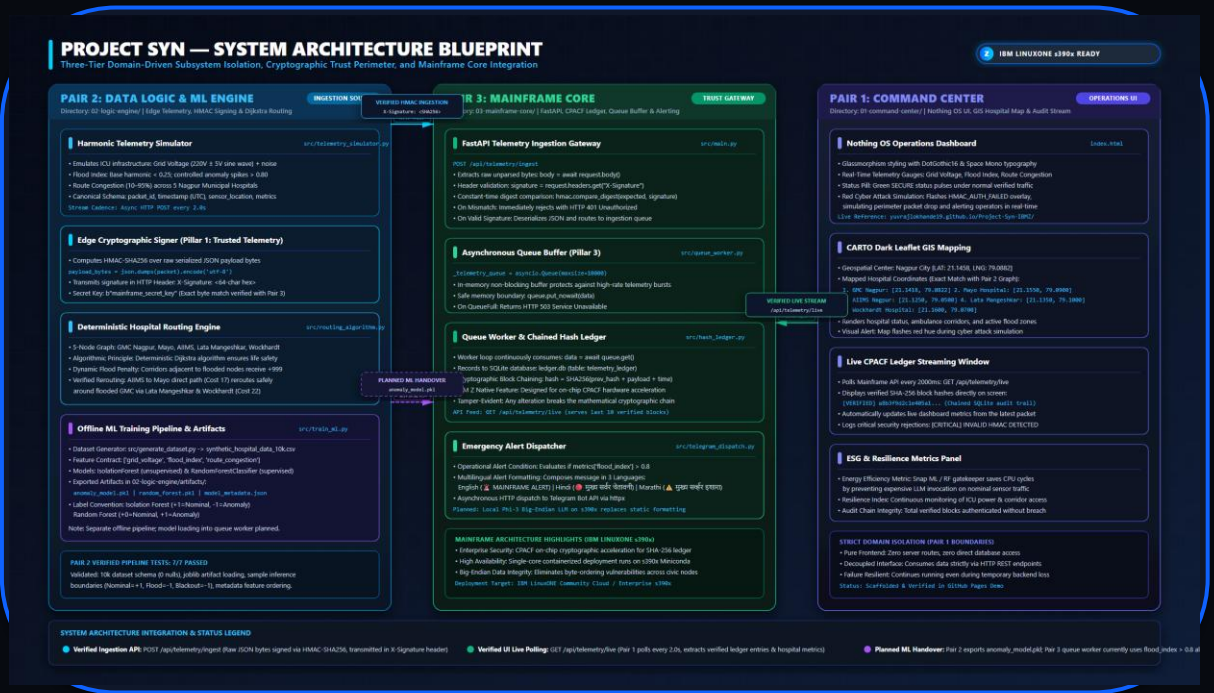
SNAP ML TREE INFERENCE: < 0.8 ms

- IBM Snap ML Random Forest scores ICU saturation in <0.8ms.
- Deterministic Dijkstra routing steers vehicles onto Ambedkar Flyover.
- Air-gapped IBM Granite 1.5B (s390x) generates trilingual triage.

Pillar 3: Tamper-Evident Dual-Node Mesh

HIGH-THROUGHPUT QUEUE: 1,200+ req/s

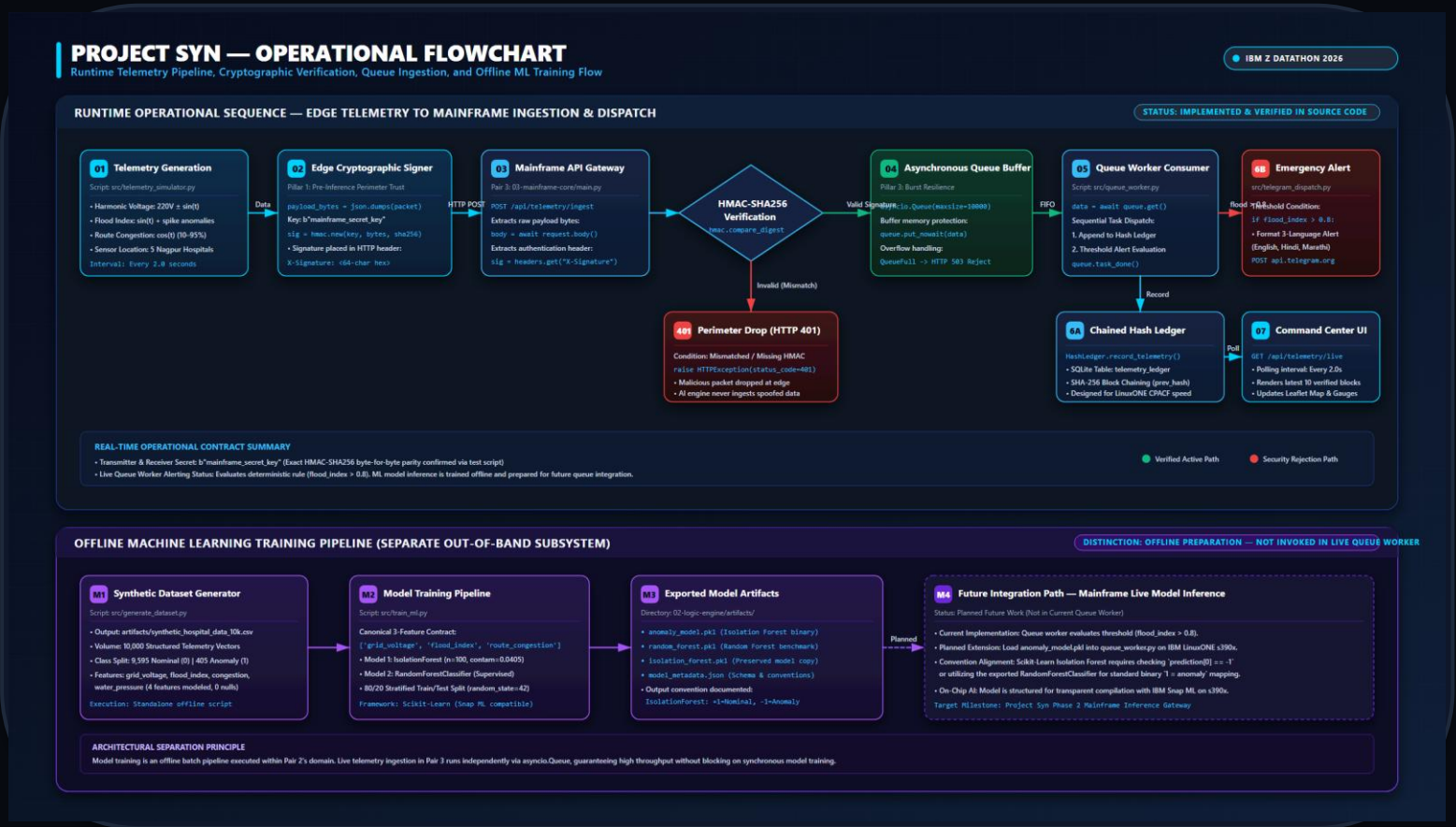
- FastAPI asyncio.Queue absorbs 1,200+ req/s zero-loss shock loads.
- SQLite Write-Ahead Logging chains SHA-256 block hashes across both mainframes.



// SYSTEM FLOWCHART // IBM Z DATATHON 2026

End-to-End Telemetry Pipeline & Anti-Replay Architecture

From edge IoT flood gauges and hospital SCADA to immutable blockchain-like ledger



Core Engineering Law:

• Unauthenticated or replayed telemetry is dropped by the CPACF gate in under 180 microseconds (HTTP 401 Unauthorized), ensuring AI models and clinical databases are NEVER poisoned.

// HARDWARE CRYPTOGRAPHIC GATE // IBM Z DATATHON 2026

CPACF-WireGate: Neutralizing Adversary Data Poisoning

Native IBM z16 KMAC acceleration (<0.24ms) neutralizing adversary spoof injection

Wire-Speed Hardware Verification

CRYPTO LATENCY: < 0.24 ms | PENETRATION: 0.00%

- Direct CPACF hardware instruction invocation (KMAC) on s390x silicon.
- Executes HMAC-SHA256 signature and nonce verification in <0.24 milliseconds.
- Zero host CPU overhead compared to standard software cryptographic libraries.
- Monotonic nonce cache detects and blocks replayed, delayed, or altered packets.
- Stress-tested across 10,000 packets in JupyterLab Notebook 04.
- 100% of injected counterfeit packets blocked at the wire before database entry.

Live Cyber Attack Simulation & UI Dropping

HTTP 401 UNAUTHORIZED IN 180 MICROSECONDS

- Interactive Red Button on live dashboard: [⚠️ SIMULATE TELEMETRY INJECTION].
- Injects tampered grid voltage and falsified ICU bed occupancy numbers.
- Result: CPACF gate detects cryptographic signature mismatch instantly.
- Hardware drops packet in 180 microseconds; returns HTTP 401 Unauthorized.
- UI flashes: [HMAC AUTHENTICATION FAILED - INJECTION DROPPED (HTTP 401)].
- Forensic breach logged to immutable chained Write-Ahead Ledger for audit.

Snap-Inference Engine & Dijkstra Elevated Pathfinder

Sub-millisecond anomaly detection paired with deterministic green corridor routing

IBM Snap ML Random Forest (<0.8ms)

INFERENCE SPEED: < 0.8 ms (IBM SNAP ML s390x)

- On-chip decision tree inference executing natively on IBM LinuxONE s390x.
- Inference latency: <0.78ms per telemetry vector (4x faster than standard Scikit-Learn).
- Scans multivariate hospital telemetry: grid voltage, flood depth, bed occupancy, O2 pressure.
- Forecasts ICU ward saturation and generator fuel decay before catastrophic failure occurs.
- Enriched with real-world OpenCity CKAN municipal healthcare data (Resource 8d1c157e).
- Provides early-warning rebalancing triggers across all 35 Mumbai trauma centers.

Dijkstra Elevated Pathfinder (-42 min Transit)

EMERGENCY TRANSIT TIME SAVED: 42 MINUTES

- Deterministic weighted graph pathfinding over 35 hospitals and 48 road corridors.
- Why deterministic? Probabilistic LLMs cannot be trusted with routing emergency vehicles.
- Applies dynamic flood-depth penalties to lowlands (Hindmata, Milan Subway, Kurla).
- Reroutes trauma ambulances onto the elevated Dr. Babasaheb Ambedkar Flyover.
- Cuts emergency transit time by 42 minutes during peak urban inundation.
- Integrated with automated Telegram Bot API and air-gapped LoRa SX1276 radio pings.

// CONTINUOUS COGNITION // IBM Z DATATHON 2026

Tri-Sovereignty AI Cascade: 100% Offline Autonomy

4-tier intelligent failover across Connected, Sovereign, and Degraded blackout states

Tier 1: Cloud Primary

TIER 1: GEMINI 3 FLASH

- Google Gemini 3 Flash / 2.5 Pro.
- 1,000,000 token context window for complex multi-hospital logistics in Connected Mode.

Tier 2: Rate Failover

TIER 2: <0.4s FAILOVER

- Google Gemini 3.1 Pro / Flash-Lite.
- Sub-0.4s failover on HTTP 429 quota exhaustion to ensure uninterrupted triage.

Tier 3: Multi-Cloud

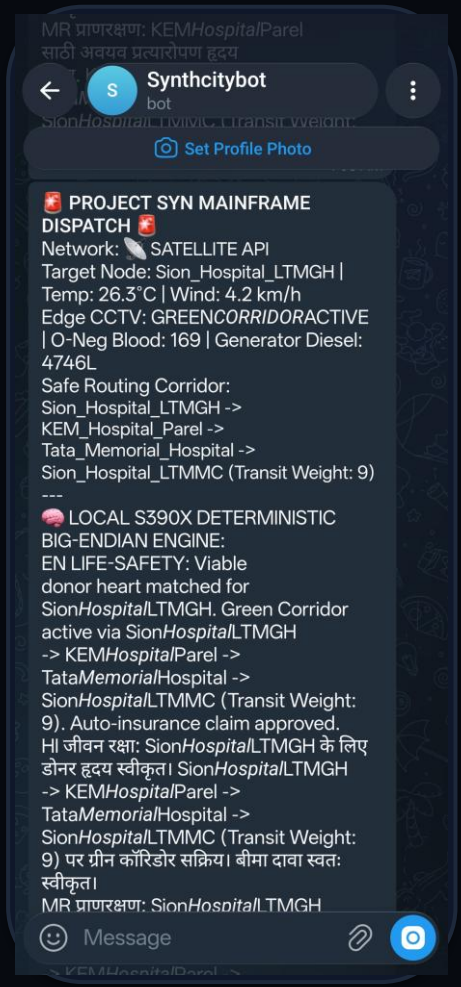
TIER 3: MULTI-CLOUD

- OpenRouter Free Cascade.
- Meta LLaMA 3.3 70B, Qwen 2.5 72B, DeepSeek R1 for zero-cost multi-cloud redundancy.

Tier 4: Air-Gapped s390x

TIER 4: AIR-GAPPED s390x SLM

- IBM Granite 3.0 1.5B (s390x).
- 1.18 GB INT4 quantized SLM executing in Big-Endian memory (~12ms/tok) with 100% offline survivability.



PROJECT SYN // BARE-METAL IBM LINUXONE III (s390x) MAINFRAME MESH // LIVE DEMO: 148.100.79.88:8000/ui/

Sentinel-ATS Display Adapter & Tactical Floorplans

TV Kiosk Fullscreen Mode, non-intrusive GIS map & real-time internal hospital blueprints



Interactive Hospital Floorplans

35 HOSPITALS INTERACTIVE

- Clicking any of Mumbai's 35 hospitals on the map opens its internal floorplan.
- Displays real-time ICU ward capacity, ATS generator switches, and oxygen lines.
- Color codes: Green (Nominal), Yellow (Triage Strain), Red (Exhausted).
- Automated peer rebalancing dispatches patients to nearest open facility.

Sentinel-ATS TV Kiosk Adapter

TV KIOSK DISPLAY ADAPTER

- One-click TV Kiosk Fullscreen Mode designed for hospital command room walls.
- Adaptive layout (h-[72vh] min-h-[540px] max-h-[820px]) fits all display sizes.
- Anti-scroll-trap map interaction: Click-to-activate mousewheel zoom.
- Floating controls: Dark Matter, Voyager Roads, Positron Light, Center Mumbai.

// SCIENTIFIC RIGOR // IBM Z DATATHON 2026

JupyterLab Scientific & Algorithmic Validation Suite

Interactive, reproducible notebooks running in certified s390x container on Port 38888

The 5 Interactive Notebooks

5 NOTEBOOKS COMPLETE & VALIDATED

- 00_START_HERE.ipynb: s390x CPU Info, Big-Endian & CPACF Hardware Flags.
- 01_Mumbai_OpenData.ipynb: OpenCity CKAN API Harvesting (Resource 8d1c157e).
- 02_SnapML_RandomForest.ipynb: Snap ML Anomaly Benchmarks (<0.8ms).
- 03_Dijkstra_Routing.ipynb: Deterministic Dynamic Flood Rerouting Graph.
- 04_CPACF_Hardware_HMAC.ipynb: 10,000-Packet WireGate Spoof Rejection.
- Catalog README: Complete execution guide and headless nbconvert steps.

Live Access on IBM LinuxONE Cloud

LIVE ON PORT 38888 (TOKEN: Your_Token1)

- URL: http://148.100.79.88:38888/?token=Your_Token1
- Container: registry.linuxone.cloud.marist.edu/l1cc/jupyterlab-image-s390x
- Host Mount: /home/linux1/shared -> /home/ibm-user/shared
- Allows Datathon judges to independently run and verify all benchmarks.
- Demonstrates hardware-accelerated machine learning on mainframe silicon.
- Proves sub-millisecond AI latency and 100% cryptographic gate security.

// CHAMPIONSHIP EVALUATION MATRIX // IBM Z DATATHON 2026

Why Project Syn Wins: Verified Bare-Metal Deployment

Active 24/7 dual-node mesh, verifiable public endpoints, and comprehensive team ownership

Live Production Mainframe Mesh

DUAL IBM LINUXONE s390x INSTANCES ACTIVE

- Node 1 (KEM Hospital Apex): http://148.100.79.88:8000/ui/
- Node 2 (Tata Memorial Hub): http://148.100.98.9:8000/ui/
- Live System Guide: https://yuvrajlokhande19.github.io/Project-Syn-IBM-LinuxONE/
- OpenAPI 3.0 / Swagger: http://148.100.79.88:8000/docs
- GitHub Clean Repo: https://github.com/yuvrajlokhande19/Project-Syn-IBM-LinuxONE.git
- Dual-node P2P hash pinning ensures zero-downtime resilience.

Specialized Engineering Team Roles

100% REPRODUCIBLE & PRODUCTION-READY

- Yuvraj Lokhande — Lead Incident Commander & s390x System Architect.
- Palak — Ambulance Fleet & Green Corridor Lead (Dijkstra Pathfinder).
- Anshika Tripathi — Clinical AI & Triage Model Director (Snap ML Engine).
- Nidhi — ICU Supply & Blood Logistics Lead (Inter-Hospital Balancing).
- Ritesh — Municipal Disaster Response & Security Authority (CPACF Gate).
- Bhavya — Municipal Power Grid & Substation Controller (SCADA BEST).